

Axioms of Probability

Motivation

- Consider repeating indefinitely a random experiment with sample space S .
- Let $E \subset S$ be an event and let $n(E)$ denote the number of times event E occurs in the first n repetitions.
- One "may" define the probability $P(E)$ of event E as the limit of relative frequency of E as $n \rightarrow \infty$ (i.e., the limit of the # of times E occurs out of n experiment trials) :

$$P(E) = \lim_{n \rightarrow \infty} \frac{n(E)}{n}$$

However, there are conceptual problems/difficulties with above definition:

- In practice, one cannot repeat experiment indefinitely
- The limit above might not exist in general.
- Even if the limit exists, there is no guarantee that it will be the same when the process is run at a later time.

→ A mathematically consistent theory is possible if we assume that $P(E)$ exists for all "measurable" subsets of S that satisfy intuitively desirable axioms.

Definition: Event Space (on σ -field)

Given a random experiment with sample space S , an event space (on σ -field or σ -algebra) $\mathcal{F}$ for S is a collection of events (subsets of S) such that:

$$(1) S \in \mathcal{F}$$

$$(2) \text{ If } E \in \mathcal{F}, \text{ then } E^c \in \mathcal{F}$$

complement of E

$$(3) \text{ If } E_1, E_2, \dots \in \mathcal{F}, \text{ then } \bigcup_{i=1}^{\infty} E_i \in \mathcal{F}$$

countable union

Remarks:

(We will only consider such events in the course
← → call them events.)

- Events in $\mathcal{F}$ are called "measurable sets" or " $\mathcal{F}$ -measurable sets".
- $\emptyset \in \mathcal{F}$ since $S^c = \emptyset$
- The union of a finite collections of events in $\mathcal{F}$ also belongs to $\mathcal{F}$ since $\bigcup_{i=1}^n E_i = \bigcup_{i=1}^{\infty} E_i$ if we set $E_i = \emptyset$ for $i \geq n+1$.
- The intersection of a finite or countably many events in $\mathcal{F}$ also belong to $\mathcal{F}$. Ex: If $E, F \in \mathcal{F}$, then $E \cap F = (E^c \cup F^c)^c \in \mathcal{F}$ by (2) and (3)
- Simplest example of $\mathcal{F}$ is $\mathcal{F} = \{\emptyset, S\}$.
- The collection of all subsets of S (power set of S) is an event space.

Definition Axioms of Probability

Given sample space S and event space $\mathcal{F}$ for S , then a real-valued function P on $\mathcal{F}$ is called a probability function if it satisfies the following:

- 1) $P(E) \geq 0$ for all $E \in \mathcal{F}$ ← $P(E)$ called probability of event E
- 2) $P(S) = 1$
- 3) If $E_1, E_2, \dots$ are mutually exclusive events in $\mathcal{F}$ (i.e., $E_i E_j = \emptyset$ for all $i \neq j$), then

$$P\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} P(E_i)$$

← "countable additivity property"

Note: We call the triplet $(S, \mathcal{F}, P)$ a probability space.

Theorem 1:

$$P(\emptyset) = 0$$

Proof:

Consider a sequence of events $E_1, E_2, \dots$ in $\mathcal{F}$ such that $E_1 = S$ and $E_i = \emptyset$ for all $i = 2, 3, \dots$.

$$\text{Then } \bigcup_{i=1}^{\infty} E_i = S$$

$$\text{and } E_i E_j = \emptyset \text{ for all } i \neq j.$$

Hence

$$P(S) = P\left(\bigcup_{i=1}^{\infty} E_i\right)$$

$$= \sum_{i=1}^{\infty} P(E_i) \quad \text{by axiom 3}$$

$$= \underbrace{P(E_1)}_S + \sum_{i=2}^{\infty} \underbrace{P(E_i)}_{\emptyset}$$

$$= P(S) + \sum_{i=2}^{\infty} \underbrace{P(\emptyset)}_{\geq 0 \text{ by axiom 1}}$$

$$\therefore P(\emptyset) = 0$$

□

Consequence of Axiom 3 - Finite Additivity

For any finite number n of mutually exclusive events

$E_1, E_2, \dots, E_n$ (in $\mathcal{F}$), we have

$$P\left(\bigcup_{i=1}^n E_i\right) = \sum_{i=1}^n P(E_i)$$

← "finite additivity"

Proof:

Use Axiom 3 by setting $E_{n+1} = E_{n+2} = \dots = \emptyset$. Then

$$P\left(\bigcup_{i=1}^n E_i\right) = P\left(\bigcup_{i=1}^{\infty} E_i\right)$$

$$= \sum_{i=1}^{\infty} P(E_i) \quad (\text{by Axiom 3})$$

$$= \sum_{i=1}^n P(E_i) + \sum_{i=n+1}^{\infty} \overset{\emptyset}{P(E_i)}$$

$= 0$ by Thm 1

□

Special Case ($n=2$):

For any mutually exclusive events E and F ,

$P(E \cup F) = P(E) + P(F)$

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Theorem 2: $P(E^c) = 1 - P(E)$

Proof: $E \cup E^c = S$ and $E \cap E^c = \emptyset$

$$\therefore 1 = P(S) = P(E \cup E^c) = P(E) + P(E^c) \quad \square$$

$\uparrow$ axiom 2 $\uparrow$ (*)

Corollary: For any event E (in $\mathcal{F}$), $0 \leq P(E) \leq 1$

Proof: Indeed

$$0 \leq P(E) = 1 - P(E^c) \leq 1 \quad \square$$

$\uparrow$ axiom 1 $\uparrow$ Thm 2 $\underbrace{\quad}_{\geq 0}$ axiom 1

Example:

A coin is called fair if H or T are equally likely in a single toss of the coin. Let us apply the probability axioms to determine $P(\{H\})$ and $P(\{T\})$:

Here $S = \{H, T\}$ and $\{H\} \cap \{T\} = \emptyset$

$\therefore$

$$1 = P(S) = P(\{H\}) + P(\{T\}) = 2 P(\{H\})$$

$\uparrow$ (*)

$$\therefore P(\{H\}) = P(\{T\}) = \frac{1}{2}.$$

$\uparrow$ equally likely assumption

$\uparrow$
This result is a direct consequence of the axioms
(and not a result of an actual experiment)

Equally Likely Outcomes

Consider a sample space $S = \{s_1, s_2, \dots, s_N\}$ with N equally likely outcomes:

$$P(\{s_1\}) = P(\{s_2\}) = \dots = P(\{s_N\})$$

Then by axioms, we directly have:

$$1 = P(S) = P\left(\bigcup_{i=1}^N \{s_i\}\right) \overset{\substack{\uparrow \\ \text{finite} \\ \text{additivity}}}{=} \sum_{i=1}^N P(\{s_i\}) \overset{\substack{\uparrow \\ \text{assumption}}}{=} N P(\{s_1\})$$

∴

$$P(\{s_i\}) = \frac{1}{N}, \quad i = 1, 2, \dots, N$$

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↑
Equally Likely Assumption imposes above
Probability function on Sample Space S .

Theorem 3:

Let S be a sample space with N equally likely outcomes,
then for all $E \subset S$,

$$P(E) = \frac{|E|}{N}$$

size of set E
(# of elements in E)

Proof:

$$P(E) = P\left(\bigcup_{s \in E} \{s\}\right)$$

$$= \sum_{s \in E} \underbrace{P(\{s\})}_{= 1/N} \quad \text{by finite additivity}$$

$$= \frac{1}{N} \sum_{s \in E} (1)$$

$$= \frac{|E|}{N} . \quad \square$$

Example: If two dice are rolled, what is the probability that the sum of the two resulting numbers is 7?

Answer: Here sample space $S = \{(i, j) : 1 \leq i, j \leq 6\} = \{1, 2, \dots, 6\}^2$

Assuming dice are fair, we have that all 36 outcomes in S are equally likely.

∴ Event E : "sum is 7" is

$$E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$$

$$\therefore P(E) = \frac{|E|}{N} = \frac{6}{36} = \frac{1}{6}$$

Example: What is the probability of getting exactly 2 Tails in 3 flips of a fair coin?

Answer:

$$S = \{HHH, HHT, HTH, HTT, TAA, THT, TTH, TTT\}$$

$$N = 8$$

$$E = \{HTT, THT, TTH\}$$

$$\therefore P(E) = \frac{|E|}{N} = \frac{3}{8}$$

$|S| = 8$

Example (where outcomes Not equally likely)

Consider a race with three competitors: A, B and C where it is assumed that A is twice as likely as B to win and C is $\frac{2}{3}$ as likely as A to win. Assuming that no ties are allowed, find the probabilities of winning for A, B and C.

Answer: Let $A = \text{"A wins"}$
 $B = \text{"B " "}$
 $C = \text{"C " "}$

Here $S = \{A, B, C\} \leftarrow$ possible winners

Know: $P(A) = 2P(B)$ and $P(C) = \frac{2}{3} P(A)$

Thus

$$\begin{aligned} 1 &= P(S) = P(A) + P(B) + P(C) \\ &= 2P(B) + P(B) + \frac{2}{3} P(A) \\ &= 2P(B) + P(B) + \frac{2}{3} (2P(B)) \\ &= \frac{13}{3} P(B) \end{aligned}$$

$$\Rightarrow P(B) = \frac{3}{13}, \quad P(A) = \frac{6}{13}, \quad P(C) = \frac{4}{13}.$$